4 (a)	Show understanding of how the stationary of one open and one close if formed via 2 diagrams	C1
	$\lambda_1 (\frac{n}{2}+\frac{1}{4}) = D$ $\lambda_2 (\frac{n+1}{2}+\frac{1}{4}) = D$ $\frac{343}{70.6}(\frac{n}{2}+\frac{1}{4}) = \frac{343}{90.8}(\frac{n}{2}+\frac{3}{4})$ $n = 3$	M1
	$D = 8.50\ m$	A1
(b)(i)	There are multiple lines of constructive interference apart from the middle line between the emitters.	M1
	The AGV might align itself to these other lines of constructive interference instead.	A1
(b)(ii)1	The intensity of signal is lower as the signal travels further spreads over a larger surface area as it is further from the AGV,	M1
	The signal is from Emitter B.	A1
(b)(ii)2	$\frac{\pi}{3}$ rad or 1.05 rad	B1
(b)(ii)3	Dist. from A to AGV = $\sqrt{7^2+50^2}$ = 50.488 m Dist. from B to AGV = $\sqrt{5^2+50^2}$ = 50.249m	C1
	$\frac{\Delta x}{\lambda} = \frac{\Delta\phi}{2\pi} = \frac{\pi}{3(2\pi)}$ $\Delta x = \frac{\lambda}{6}$ Path Difference = $\frac{\lambda}{6}$ = 50.488 – 50.249 = 0.238m	

	$\lambda = 1.429$ m	C1
	$f = \dfrac{3 \times 10^8}{1.429}$	C1
	$= 2.10 \times 10^8$ Hz (2 to 3 sf)	A1

5 (a)	Electric potential at a point is the **work done per unit positive**	B1